

PS5 Writeup

Part A

What difference does choosing a specific day to plot the data for versus calculating the yearly average have on our graphs (i.e., in terms of the r^2 values and the fit of the resulting curves)? Interpret the results.

Consider the temperature data for a single day against time (Fig. 1). There is no particularly obvious trend with a large amount of white noise. The linear regression fit is slightly linear but with a very weak correlation ($r^2 = 0.05$). The yearly averaged temperature (Fig. 2) shows more trend in the line however still with very weak correlation of $r^2 = 0.19$.

Why do you think these graphs are so noisy? Which one is more noisy?

Single day data has lower correlation which indicates more noise. It is not surprising the data is noisy considering the possibly large day to day variations in temperature. Averaging over the year removes day to day noise (which leads to higher correlation coefficient values) although it is clear there is other factors which affect temperature variation such as variation in geography and long-term weather phenomena.

How do these graphs support or contradict the claim that global warming is leading to an increase in temperature? The slope and the standard error-to-slope ratio could be helpful in thinking about this.

Based on the low values of the error-to-slope ratio in both fits, this suggests the fit line is hugely uncertain when applied to the data. The year-average data is slightly less uncertain. However both are not good enough to accept any form of trend. One would say that the data does not 'support' the claim global warming leads to increasing temperature, however only in the sense that there is no conclusive evidence for the claim and not that there is evidence of its contradiction (we only fail to reject the notion the data appears random without trend).

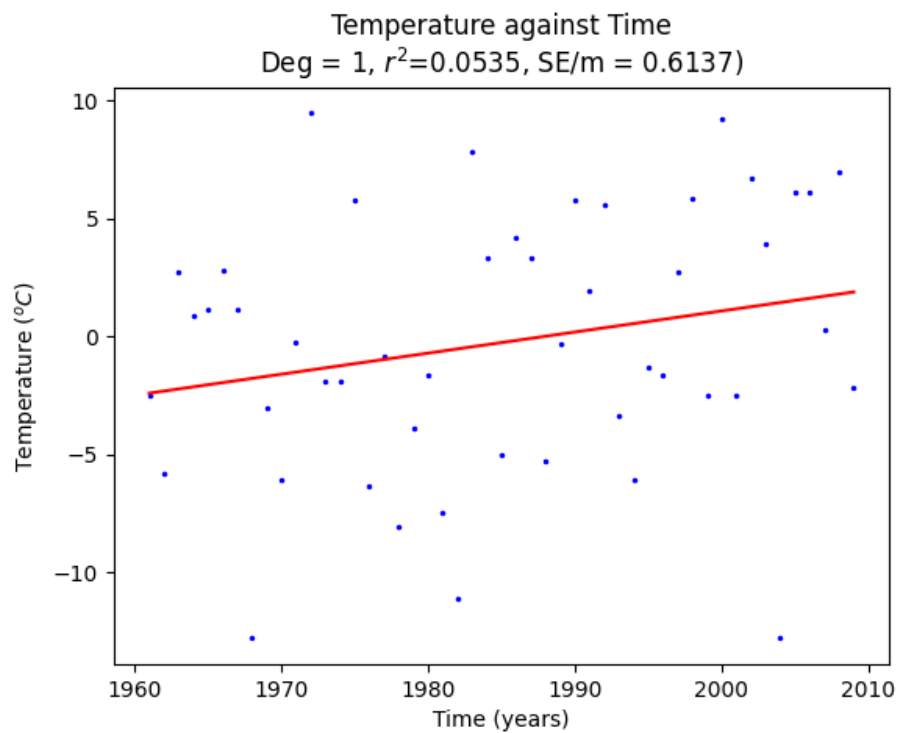


Figure 1

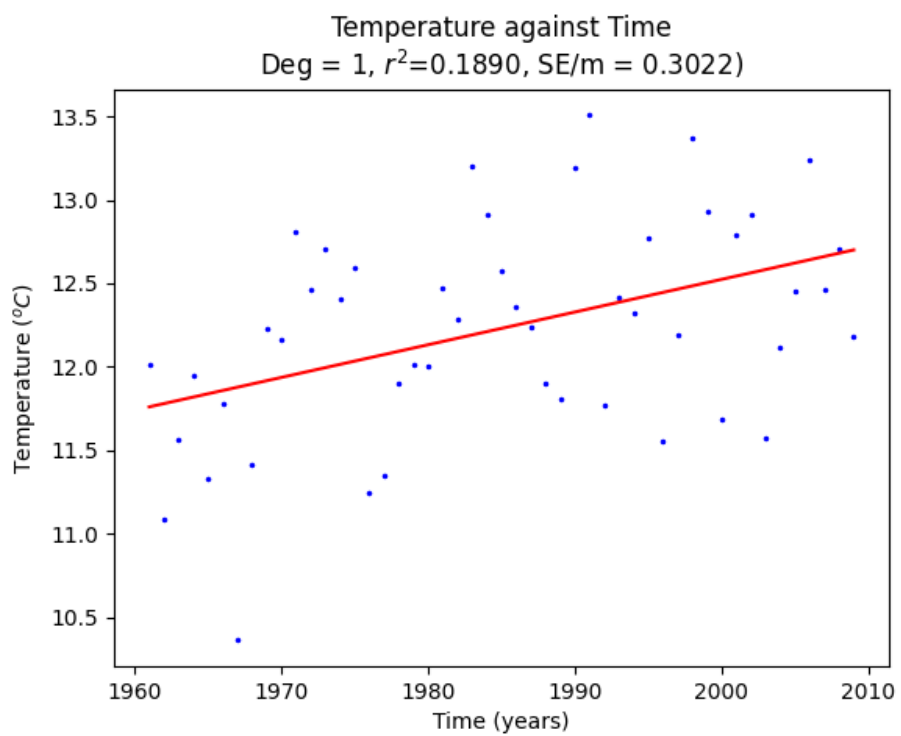


Figure 2

Part B

How does this graph compare to the graphs from part A (i.e., in terms of the r^2 values, the fit of the resulting curves, and whether the graph supports/contradicts our claim about global warming)? Interpret the results.

Qualitatively the data of temperature averaged over each year and multiple cities (Fig. 3) qualitatively matches a trend much better than the data of the previous graphs. This is confirmed quantitatively with a r^2 -value of 0.75 which can be considered a strong correlation - which is further backed by a small error-to-slope ratio indicating the trend is pretty significantly positive. The results therefore support the claim about global warming.

Why do you think this is the case?

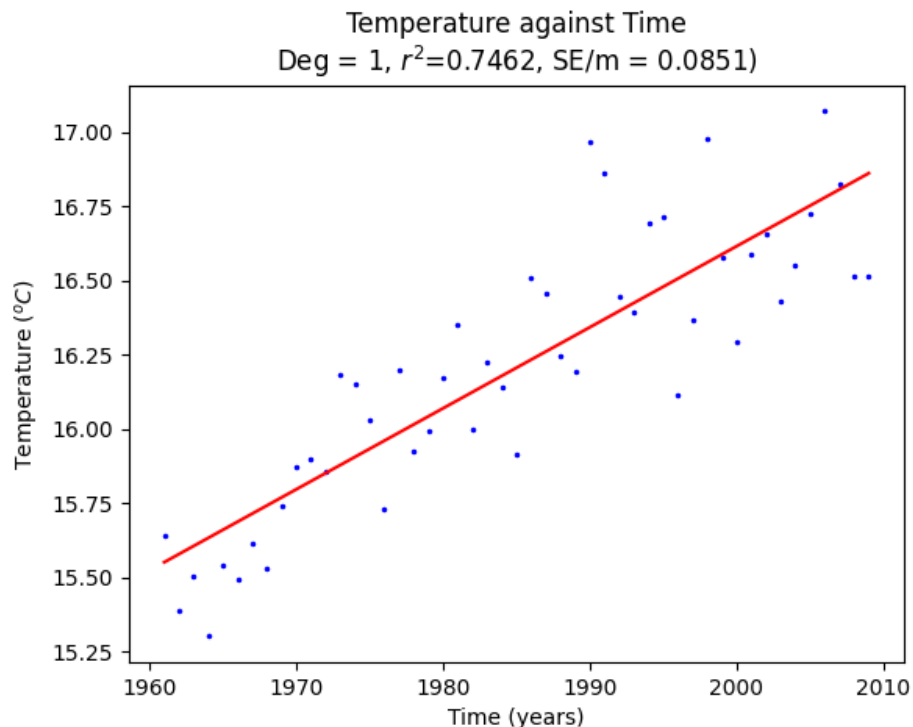
It seems the data by averaging over multiple cities, smoothens out the noise associated with variation in geographic locations. As a result, the trend in data is much clearer and not obscured by noise.

How would we expect the results to differ if we used 3 different cities? What about 100 different cities?

Averaging over 3 cities would likely not be enough samples to reduce variation, averaging over 100 cities could reduce variation even more, however it is almost certainly the case there would be still variation left over from other sources.

How would the results have changed if all 21 cities were in the same region of the United States (for ex., New England)?

The entire purpose of averaging over cities is to reduce the variation of temperature dependent on the noise introduced by spatial variation - it is not the cities themselves which are 'noisy' rather the geographic location they are. Therefore sampling cities over the same region means there is a systematic bias to only smoothing out temperature variations on a local geographic scale - which are likely small - and not a larger scale so we would see larger noise still.



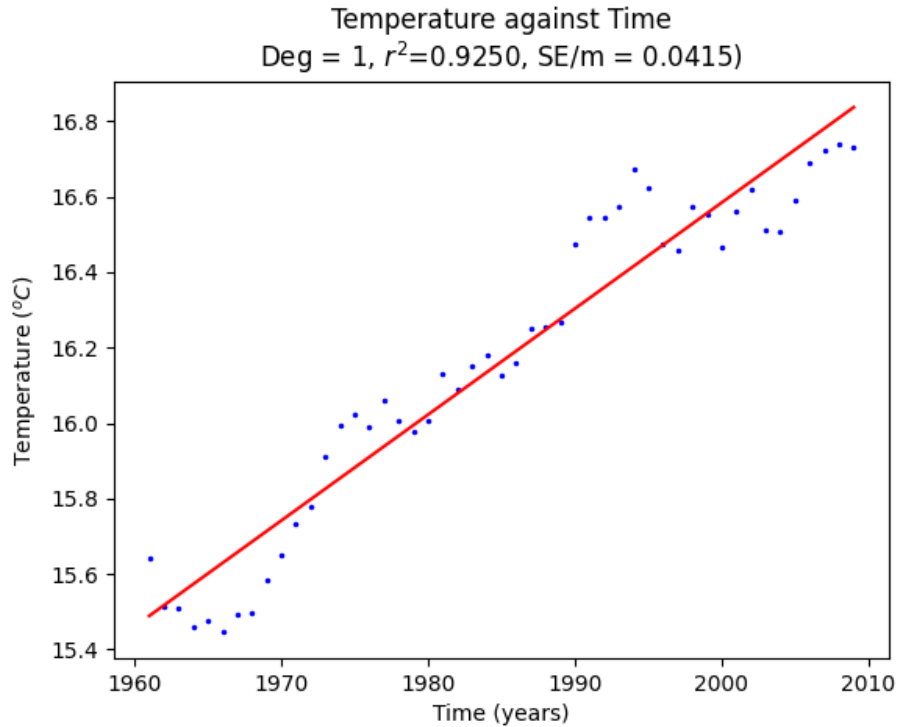
Part C

How does this graph compare to the graphs from part A and B (i.e., in terms of the r^2 values, the fit of the resulting curves, and whether the graph supports/contradicts our claim about global warming)? Interpret the results.

The graph (Fig. 4) shows even better support for the claim on global warming with very strong correlation (0.93) and even lower error-to-slope ratio. There is a qualitative difference in this graph compare to the rest - where as in previous graphs, residuals seem to be the result of random noise, there is a clear underlying trend to the residuals of Fig. 4 which indicates underlying temporal behavior which a linear regression is not capturing. Nevertheless, such a trend still appears upwards and does not interfere with our claim.

Why do you think this is the case?

In our moving average, we now add together and take into account all the forms of variation - day to day, geographic as well as any longer term noise which has a timescale spanning over a year. This allows the trend to be sharper than all for all previous datasets. Because there is a genuine different behaviour, this is now visible as it is not obscured by all the noise.



Part D. I

How do these models compare to each other?

In comparing the models as seen by Fig. 5, 6 and 7, as the degree of the polynomial increases, the models 'fit better' from the point that r^2 -values decrease. The linear and quadratic models fit similarly with the quadratic model only introducing a slight bend. The 20-degree polynomial has much different shape which appears to model the unknown variational behaviour discussed in Part C.

Which one has the best r^2 ? Why?

The higher the degree, the better the r^2 . This is simply because the residuals are much smaller as it is easier for larger number of parameter to match the observed values, and so from the perspective of the functional form, there is less 'noise', but the linear and quadratic regression cannot be so flexible so only fit to general 'trends' in the data.

Which model best fits the data? Why?

From the perspective of r^2 , the higher the degree the better the model fits. However the r^2 metric can be misleading as it only considers the fit to observational results which are seen. Excluding r^2 , it is not particularly obvious which fit is better without other tests, errors and data.

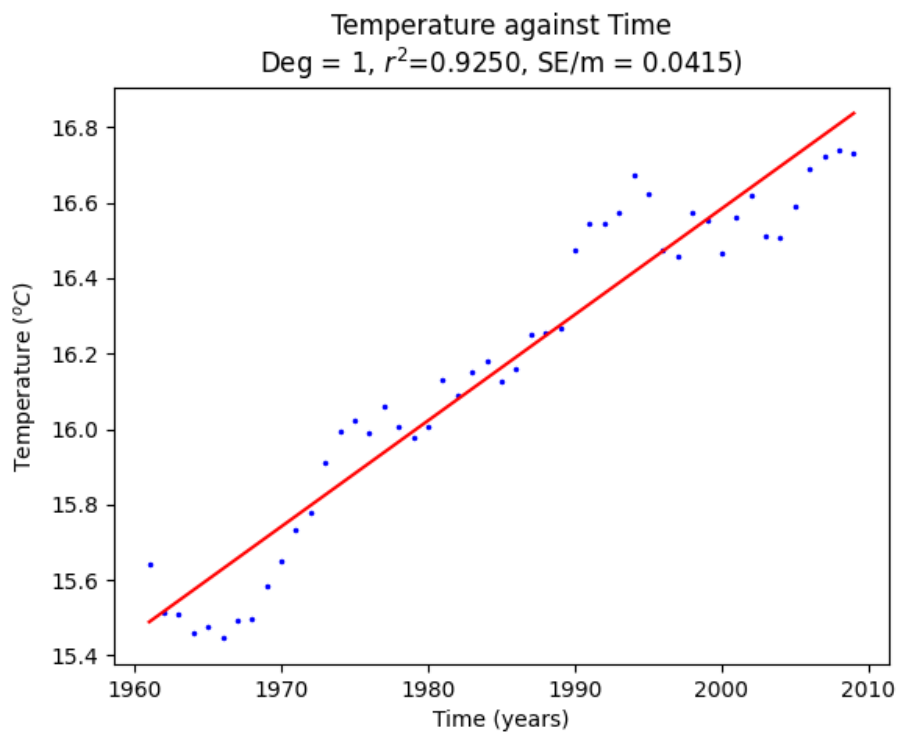


Figure 5

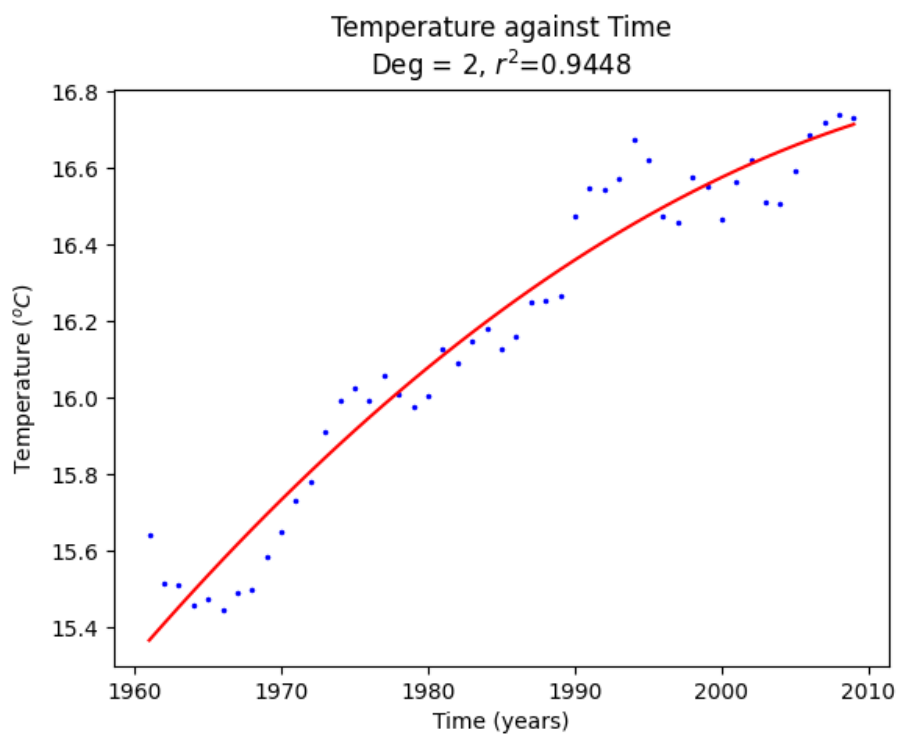


Figure 6

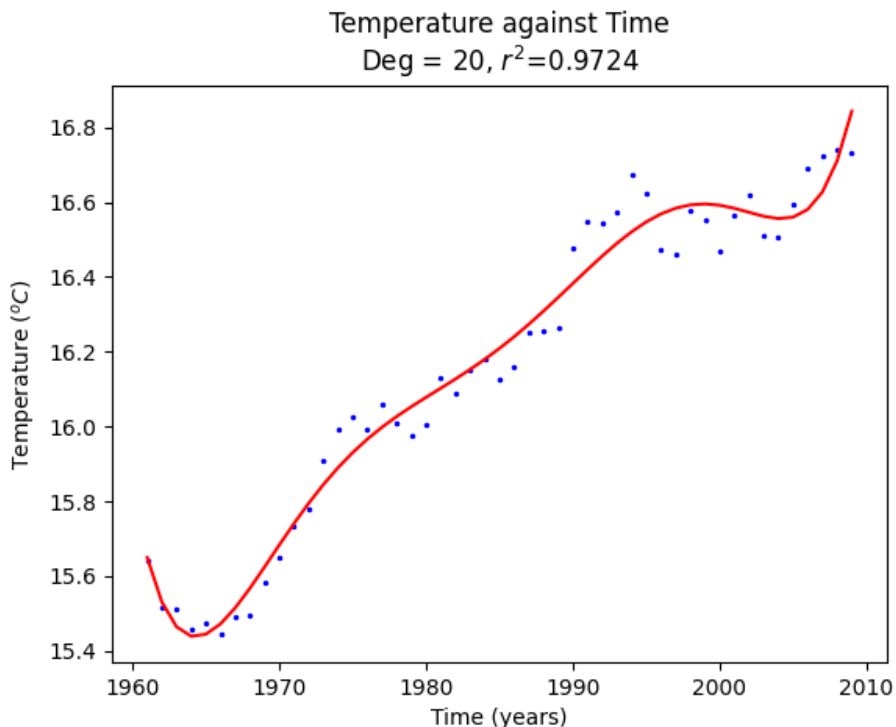


Figure 7

Part D. II

How did the different models perform? How did their RMSEs compare?

The results are reversed compared to Part D. I, with the linear model performing better than the quadratic and 20-degree (the smaller the RMSE the better). This reveals the quadratic and 20-degree are not the correct fits to the data and are clearly overfitting to the training data.

Which model performed the best? Which model performed the worst? Are they the same as those in part D.2.I? Why

Linear model performs the best, 20-degree performs the worst. It seems that the 20-degree captures the fine structure behaviour of the training set data, however in doing so ends up neglecting the general trends and as a result poorly extrapolates outside the training regions. Since the linear model only picks up the general trend, it ends up roughly estimating where the extrapolated points would be and is relatively close to that. One could say the data is still predominantly linear in nature.

If we had generated the models using the A.4.II data (i.e. average annual temperature of New York City) instead of the 5-year moving average over 22 cities, how would the prediction results 2010-2015 have changed?

If more noise is introduced, it is likely the 20-order polynomial fits to the noise and performs even

worse in prediction. Noise could potentially straighten out the quadratic as there the trend is less obvious which could make performance become more similar to the linear model.

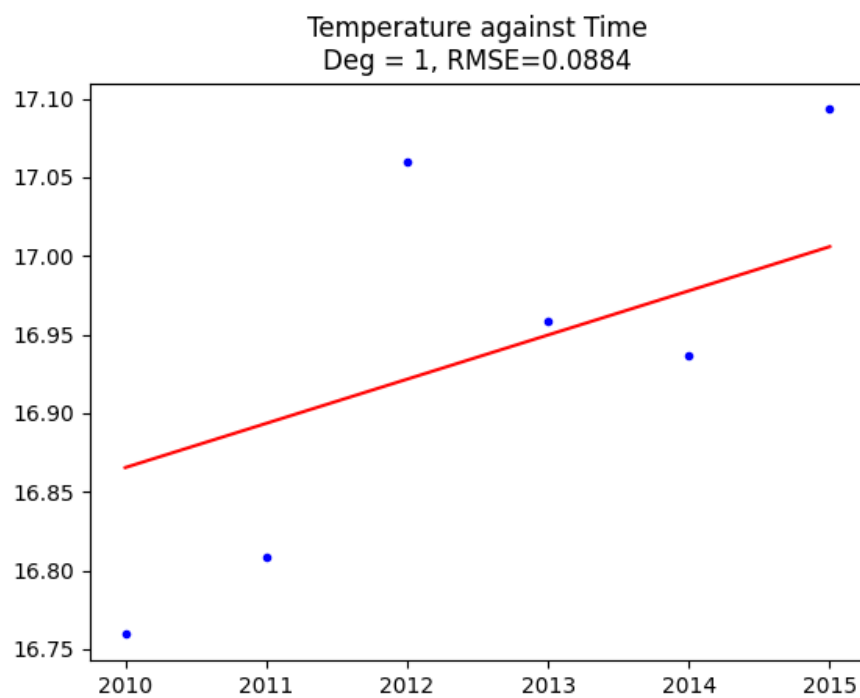


Figure 8

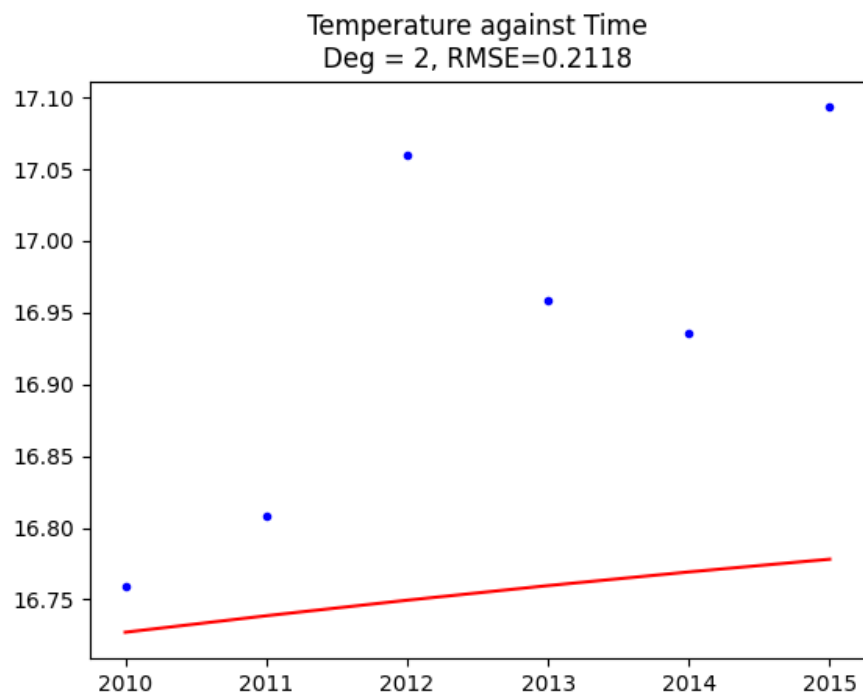


Figure 9

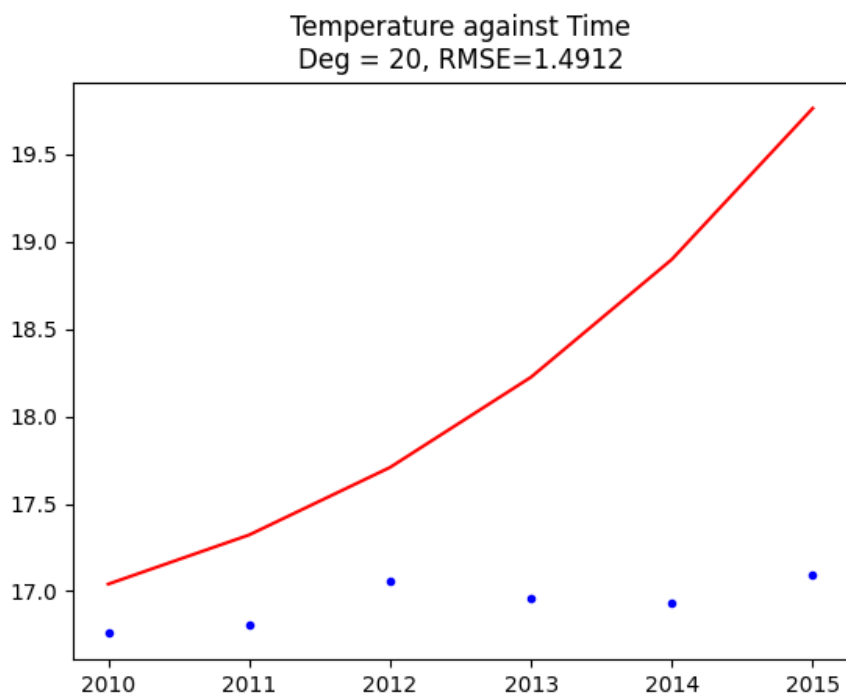


Figure 10

Part E

Does the result match our claim (i.e., temperature variation is getting larger over these years)?

The result shows poor trend, there is enough variation that there is no support for the claim (weak correlation $r^2 = 0.3$) and fairly high error-to-slope ratio. Moreover, if anything the trend is suggesting the opposite that variation is reducing which would contradict the claim. The data shows an inconclusive result.

Can you think of ways to improve our analysis?

We should look to reduce variation due to noise. In computing the standard deviation, we average over each day temperature for the given list of cities in a given year. Single day temperatures across cities may still have large variation, we may see less noise if a moving average is applied to temperatures at a day of a given city and given year. The inclusion of more cities would also contribute as well as possibly studying a larger range of years.

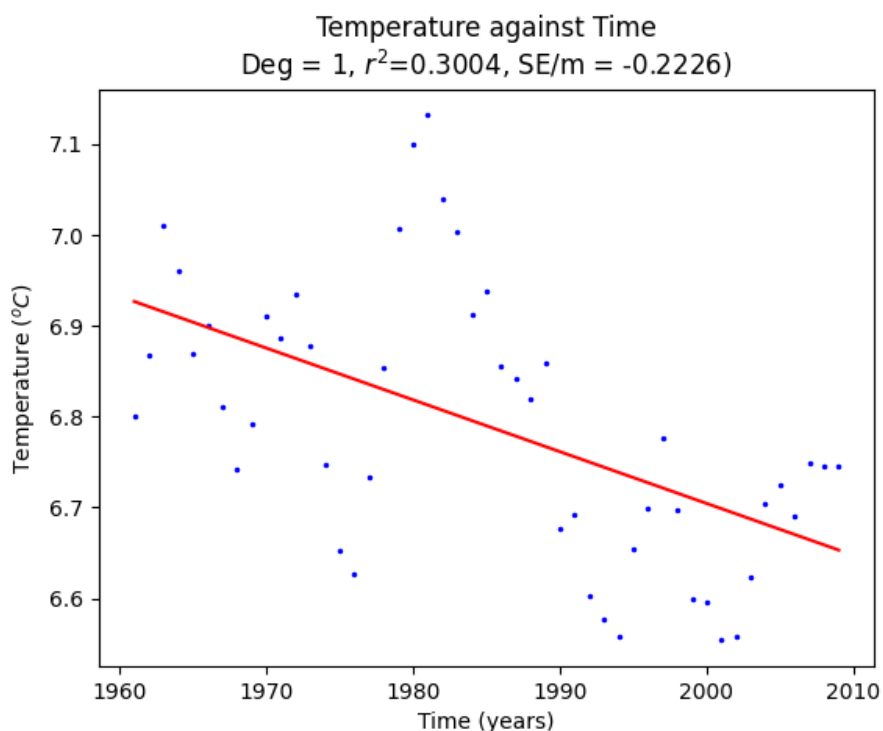


Figure 11